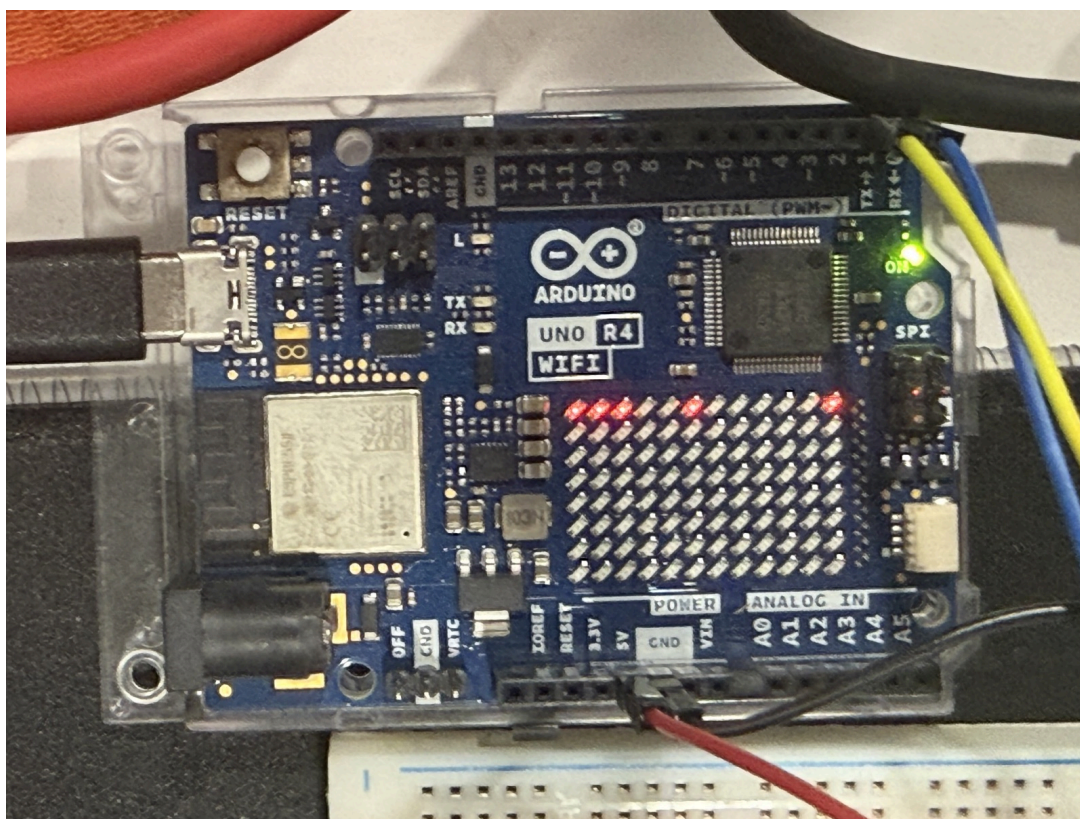


Reading the Uno LED display

The Uno R4 WiFi has a **12-column × 8-row red LED matrix**. It gives a quick view of connectivity and activity while you play. The green **ON** light and the small board **TX/RX** lights in the photograph are separate board LEDs, not pixels in the red matrix.

Evidence: the photograph below shows the owner's powered Uno on 2026-09-23. The meanings and examples on this page are **source-reviewed** against the current Uno firmware. A still photograph cannot establish which of the momentary activity or heartbeat pixels were on at a particular instant. A targeted two-board check confirmed view selection and internal envelope/VU values for a browser note. The owner then confirmed that a held key lights a Status voice bar, flashes column 10, and produces movement in VU and Scope. Recent two-board tests also confirmed Pico-forwarded wireless notes and disconnect releases; the physical DIN input and Uno audio output remain untested. [Matrix implementation](#), [hardware test log](#).



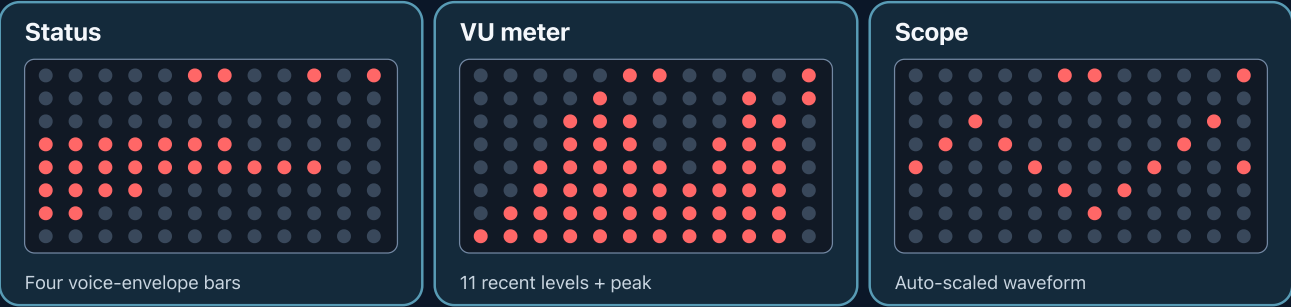
Live photograph of the earlier Uno-network build. The current top row uses those former network positions for sound and performance status. The Pico handles Wi-Fi and RTP-MIDI connections.

Choose a view

In the browser panel, open **Explore** or **All controls**, find **LED matrix**, and choose **Status**, **VU meter**, or **Scope**. These change only the Uno display; they do not change the sound. Status is the default after boot or a synth reset. The selector sends Pico-to-Uno Program Changes **10**, **11**, and **12** respectively. The Uno scrolls **Viz: Status**, **Viz: VU**, or **Viz: Scope** across the matrix while note processing continues, then returns to the selected display. These display commands are accepted only on the Pico UART, not on the separate DIN or RTP-MIDI inputs. [Browser control](#), [Uno display selection](#).

Uno R4 LED matrix views

Same 12x8 display · top row carries sound, performance, link, and timing indicators



Illustrative frames, not captures. Red pixels are lit LEDs; positions change while the Uno runs.
VU bars are recent mono levels over time, not frequency bands or left/right channels.

Illustrative frames for the current gateway build, not captures. Lit pixels vary with Pico traffic, notes, waveform, and the heartbeat.

View	What the lower rows show	Useful when
Status	Up to four horizontal bars, one for each synth voice. Each bar follows that voice's envelope level.	Checking which voices are active or releasing.
VU meter	Eleven vertical bars show recent mono output levels, oldest at left and newest beside the right-edge peak dot. The levels come from the synthesized signal before the DAC.	Seeing how loudness changes across roughly the last third of a second.
Scope	A moving, automatically scaled trace with one dot per column. It samples the synthesized output before the DAC.	Seeing the shape and movement of the generated waveform.

The VU bars are **recent levels over time**. They are not frequency bands or stereo channels; QuarkWave currently produces one mono signal. Each settled VU frame adds one level about every 33 ms. The rightmost column holds the peak dot and leaves its bottom pixel free for the timing diagnostic. **VU** and **Scope** are visual guides, not calibrated level meters or measurements at the audio jack. Because the audio output stage is not assembled yet, their real-world relationship to a mixer or headphones remains untested. [Output sampling and display calculations](#), [connection guide](#).

The top row stays useful in every view

Count columns **1–12 from left to right as viewed on the board**. The firmware draws these indicators on the top row in all three settled views. Boot and scrolling mode labels temporarily replace the view, including its top row. A pixel marked “brief” is expected to flash and go dark; it does not represent a persistent fault.

Uno R4 LED matrix · top row

Columns 1–12, left to right as viewed on the board · illustrative moment



Red = lit in this example · gray = unlit · FLASH indicators remain lit briefly after an event.

- **1–6:** 1 Pico sound loaded · 2 external MIDI used · 3 external clock active · 4 sustain · 5 arpeggiator enabled · 6 Pico link
- **7–12:** 7 Pico MIDI RX · 8 voice stolen · 9 external sound activity · 10 note · 11 audio timing slip · 12 heartbeat

Column	Meaning in this build	When it lights
1	Pico sound loaded	The Uno acknowledged applying a Pico patch. This does not prove the sound is unchanged afterward.
2	External MIDI used	A DIN or Pico-forwarded wireless note or sound control has been accepted since boot or the last Pico patch sync. Columns 1 and 2 can both be on. Clock alone does not light it.
3	External clock active	External tempo is selected and a valid, recent DIN or Pico-forwarded RTP clock stream is the selected source. It goes dark when pulses stop for about two seconds; the last tempo is retained. Transport Start alone is not a valid clock stream.
4	Sustain held	The Uno's sustain state is on.
5	Arpeggiator enabled	An arpeggiator mode other than Off is selected. This does not mean it is currently stepping: transport may be stopped or no keys may be held.
6	Pico link	While the Uno has received a Pico readiness request within roughly 3.5 seconds. This is a handshake indicator, not merely a powered UART.
7	Pico MIDI receive, brief	For about 200 ms after the Uno handles a Pico UART note, sound control, Program Change, or SysEx. Pico readiness requests can flash this pixel even when nobody is playing. Clock and transport do not light it.

Column	Meaning in this build	When it lights
8	Voice stolen, brief	For about 400 ms when a new note needs a voice and all four are occupied. The Uno reuses one voice according to its selected voice-steal rule.
9	External sound activity, brief	For about 300 ms after a DIN note, CC, or pitch bend, or a Pico-forwarded wireless activity marker, note, or sustain message. This shows activity, not an RTP connection. MIDI Clock and transport do not light it. A DIN sound Program Change or sound SysEx can change the sound without flashing this pixel.
10	Note started, brief	For about 400 ms after a synth note starts, in all three settled views.
11	Audio timing slip, brief	For about 700 ms when the audio scheduler finds itself more than 2 ms behind and skips stale scheduled samples. This reports a scheduling event, not a measured DAC underrun count or an audible glitch.
12	Heartbeat	Toggles about every 250 ms; a regular blink shows the display loop is updating.

The **bottom-right pixel** is a separate display timing diagnostic. It can light for a frame if more than 200 ms elapsed between LED updates, such as after a blocking operation. It does not measure the audio scheduler; column 11 does that. An occasional flash alone does not identify its cause. [Top-row and diagnostic overlays](#).

What happens at startup

The matrix first plays a boot animation and scrolls **QuarkWave**. The Uno no longer starts Wi-Fi or scrolls an IP address or RTP connection label; the optional Pico owns those network functions. Changing the display view scrolls **Viz: Status**, **Viz: VU**, or **Viz: Scope**. The selected view resumes after the text finishes crossing the matrix and a brief hold. Note processing continues during the scroll. [Boot animation and view labels](#).

Quick checks

- **Heartbeat blinks, Pico pixel is dark:** the display loop is running, but the Uno has not received a recent Pico readiness request. Check the browser's separate Pico and Uno connection labels and the inter-board wiring. The Pico-to-Uno link can be offline while the Uno remains playable from its separate DIN input.
- **Column 9 flashes:** a DIN sound message or Pico-forwarded wireless sound message arrived. Check the Pico's RTP connection label to distinguish wireless session status; column 9 by itself cannot do that.
- **Column 8 flashes:** a fifth overlapping note needed one of the four voices; shorten releases or play fewer overlapping notes if this is unwanted.
- **Column 11 flashes repeatedly:** the audio loop is falling behind its schedule. Try fewer voices or lighter effects and record the patch and playing conditions for a performance test. An isolated flash does not prove an audible glitch.
- **Columns 1 and 2 are both lit:** the Pico loaded a sound, then external MIDI was used. Column 2 does not distinguish playing notes from changing sound controls.
- **Voice bars move but VU seems still:** switch to VU and play again. Status follows individual voice envelopes; VU follows the final synthesized output. A new VU view starts with an empty history and fills as frames arrive.

- **A mode label is scrolling:** wait for the text to cross the matrix and the selected view to resume. Keys remain playable during the scroll.

These are reading aids, not a substitute for checking MIDI reception and audio on the assembled instrument. [Pico–Uno connection behavior](#), [hardware test log](#).